

(103)

Summary

- * Vector form is $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$
- * It can also be represented as $\langle x(t), y(t), z(t) \rangle$
- * $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$
 $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$
- * Domain of a vector valued function
vector function, $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ is
the Domain of the components.
 $\therefore$ Domain of $\vec{r}(t)$ = Intersection of domain
of its components.

- * Tangent line: Equation of tangent line is

$$\vec{r}(t) = \vec{r}(t_0) + t \vec{r}'(t_0)$$

- * Velocity, speed, Acceleration:-

$$\text{Speed} = \|\vec{r}'(t)\|, \text{Velocity} = \frac{d\vec{r}}{dt}, \text{Acceleration} = \frac{d^2\vec{r}}{dt^2}.$$

- * Displacement and Distance travelled:-

$$\text{Displacement} = \int_{t_1}^{t_2} \vec{r}'(t) dt = \vec{r}(t_2) - \vec{r}(t_1)$$

$$\text{Distance travelled} = \int_{t_1}^{t_2} \|\vec{r}'(t)\| dt$$

- * The del operator (∇)

$$\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

$$\text{Gradient} = \nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

* Laplacian operator (∇^2)

$$\nabla^2 = \nabla \cdot \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\Rightarrow \nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

* Divergence and Curl

If the vector field is given by

$$\vec{F} = f(x, y, z) \hat{i} + g(x, y, z) \hat{j} + h(x, y, z) \hat{k}, \text{ then}$$

$$\text{div } \vec{F} = \nabla \cdot \vec{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

$$\text{Curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix}$$

* $\nabla \cdot \vec{F}$ acts normal to the surface $f(x, y, z) = C$

* $\nabla \cdot \vec{F} = 0 \rightarrow$ Solenoidal

* $\nabla \times \vec{F} = 0 \rightarrow$ Irrotational

* f increases most rapidly in the direction of ∇f .

* f decreases most rapidly in the direction of $-\nabla f$

* Rate of Change of f at (a, b) is $\|\nabla f\|_{(a, b)}$

(105)

* Conservative Field :- $\vec{F}$ is said to be conservative if $\vec{F} = \nabla \phi$

Also $\nabla \times \vec{F} = 0 \Rightarrow \vec{F} = \nabla \phi$

* Line Integral :- $\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \|\vec{r}'(t)\| dt$

* Line Integral of Vector Field where $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$ in 2D

Also $\int_C \vec{F} \cdot d\vec{r} = \int_C f(x, y) dx + g(x, y) dy$ where

$\vec{F} = f(x, y)\vec{i} + g(x, y)\vec{j}$ and $\nabla \cdot \vec{F} = \text{div}$

$d\vec{r} = dx\vec{i} + dy\vec{j}$ in 2D

* Work done :- $W = \int_C \vec{F} \cdot d\vec{r}$

* Fundamental Theorem of line Integral

$\int_{(x_0, y_0)}^{(x_1, y_1)} \vec{F} \cdot d\vec{r} = \phi(x_1, y_1) - \phi(x_0, y_0)$ where $\vec{F}$ is a conservative field (ie $\vec{F} = \nabla \phi$)

① Potential function is ϕ and

② Potential energy is $-\phi$.